

# Technical Report: Appendices

SeaNergy | Hyper Grey | PS 26058

Companion to DOC-TR-001. Referenced from the main report; not counted against its page budget.

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## A The candidate grid in full

The solver evaluates the Cartesian product of centre frequency, bandwidth fraction and pulse duration. Amplitude is tied to duration, so the grid is three-dimensional rather than four.

| Axis               | Points     | Values                                                                                        |
|--------------------|------------|-----------------------------------------------------------------------------------------------|
| Centre frequency   | 13         | 148.0, 173.3, 198.7, 224.0, 249.3, 274.7, 300.0, 325.3, 350.7, 376.0, 401.3, 426.7, 452.0 kHz |
| Bandwidth fraction | 6          | 0.90, 0.70, 0.50, 0.32, 0.18, 0.08 of centre                                                  |
| Pulse duration     | 6          | 0.5, 2.8, 5.1, 7.4, 9.7, 12.0 ms                                                              |
| Amplitude          | tied       | 0.92, 0.82, 0.72, 0.62, 0.52, 0.42 of full scale                                              |
| <b>Total</b>       | <b>468</b> | $13 \times 6 \times 6$                                                                        |

**Table 1:** *The candidate grid, water configuration.*

### A.1 Why only nine centres are ever selected

Bandwidth is clipped so the sweep stays inside the transducer band:  $B \leq 2 \min(f_c - f_{lo}, f_{hi} - f_c)$  and separately  $B \leq 0.9f_c$ . Together those make achievable bandwidth peak at 350.7 kHz and fall away on either side.

| Centre (kHz) | Max $B$ (kHz) | Centre (kHz) | Max $B$ (kHz) |
|--------------|---------------|--------------|---------------|
| 148.0        | 96.0          | 325.3        | 292.8         |
| 173.3        | 146.7         | <b>350.7</b> | <b>298.7</b>  |
| 198.7        | 178.8         | 376.0        | 248.0         |
| 224.0        | 201.6         | 401.3        | 197.3         |
| 249.3        | 224.4         | 426.7        | 146.7         |
| 274.7        | 247.2         | 452.0        | 96.0          |
| 300.0        | 270.0         |              |               |

**Table 2:** Achievable bandwidth by centre frequency. The four centres above 350.7kHz offer less bandwidth and suffer more absorption, so they are strictly dominated and never win. They remain in the grid because the same construction serves the bench configuration, where the band edges differ and the peak sits elsewhere.

## B Selection rule, stated precisely

$$\begin{aligned} \text{margin}(N) &= 3 + \min(12, N/70) \quad \text{dB} \\ \text{NL}(N) &= 52 + \min(8, N/140) \quad \text{dB} \\ \text{feasible} &= (\text{SNR} \geq \text{DT} + \text{margin}) \wedge (E \leq E_{\max}) \end{aligned}$$

If no candidate is feasible the constraint relaxes to  $\text{SNR} \geq \text{DT}$ . If that set is also empty the candidate with the greatest predicted SNR is selected and the decision is flagged infeasible, so the mission is told the request cannot be met rather than being handed a pulse that appears ordinary. Ties are broken on widest bandwidth first, then lowest energy.

Mode and window follow from the achieved margin  $m = \text{SNR} - \text{DT}$ :

| Condition              | Mode             | Window       |
|------------------------|------------------|--------------|
| $m < 2 \text{ dB}$     | Barker-13        | Hann         |
| $m < 3 \text{ dB}$     | LFM or geometric | Hann         |
| Depth $> 90 \text{ m}$ | Geometric        | by turbidity |
| $N > 400 \text{ NTU}$  | by margin        | Blackman     |
| Otherwise              | LFM              | Hamming      |

**Table 3:** Mode and window selection.

## C Complete uncertainty budget

Sonar range uncertainty at 10 m is 7.4 mm standard, against 5.8 mm for the tape measure serving as the reference. The two are the same order, which is the correct relationship: the system is not being validated far beyond the precision of the instrument validating it.

## D Bill of materials by subsystem

## E Test index and raw data

Every test identifier used in the report resolves to a result entry and a raw data file under 06-validation/.

## F Firmware pin allocation

| Quantity                  | Primary contributors                         | Standard           | $k = 2$            |
|---------------------------|----------------------------------------------|--------------------|--------------------|
| Frequency at 40 kHz       | Timebase 50 ppm, transform interpolation     | 18 Hz              | 36 Hz              |
| Spectral level            | Window correction, amplitude accuracy, noise | 0.30 dB            | 0.60 dB            |
| Total harmonic distortion | Harmonic-bin integration, noise subtraction  | 0.04 pt            | 0.08 pt            |
| Current below 2 mA        | Shunt tolerance, offset, quantisation        | 0.07 mA            | 0.14 mA            |
| Current 20–100 mA         | Shunt tolerance and gain                     | 0.65 %             | 1.30 %             |
| Complete-ping energy      | Voltage, current and timing propagation      | 2.1 %              | 4.2 %              |
| Water temperature         | Probe accuracy, spatial gradient             | 0.29 °C            | 0.58 °C            |
| Tape range at 10 m        | Reading, alignment, target plane             | 5.8 mm             | 11.6 mm            |
| Sonar range at 10 m       | Sound speed, peak interpolation, timing      | 7.4 mm             | 14.8 mm            |
| Turbidity                 | Sensor repeatability and settling            | 3.2 % +<br>0.6 NTU | 6.4 % +<br>1.2 NTU |
| Salinity preparation      | Scale resolution, tank volume                | 0.012 ppt          | 0.024 ppt          |
| Enclosure mass            | Scale resolution, surface water removal      | 0.10 g             | 0.20 g             |

**Table 4:** Full uncertainty budget, root-sum-square propagation. Range uncertainty uses  $R = ct/2$  with the temperature-derived sound-speed term and the round-trip timing term treated as independent.

## G Reproducing every figure and number

The dataset regenerates byte for byte from seed 20,260,829:

7d5d50dce7b92d3b083f80b5b52510589e6372d6ab87e78e8686e56cfb126053

| Ref                         | Part                          | Function                              | Qty |
|-----------------------------|-------------------------------|---------------------------------------|-----|
| <i>Power and protection</i> |                               |                                       |     |
| F1                          | MF-MSMF200X-2                 | 2.0 A resettable fuse                 | 1   |
| D1                          | SMBJ15A                       | 15 V transient suppressor             | 1   |
| Q1                          | DMP3010LK3-13                 | Reverse-polarity blocking             | 1   |
| U1                          | LM2596S-ADJ                   | Switching pre-regulator               | 1   |
| U2                          | AP2112K-3.3                   | 3.3 V low-dropout regulator           | 1   |
| BAT1                        | 3S-2600-LIPO                  | Protected battery pack                | 1   |
| <i>Control and transmit</i> |                               |                                       |     |
| U3                          | ESP32-S3-DevKitC-1-N8         | Controller                            | 1   |
| U5                          | SN74LVC16373DLR               | Parallel bus latch                    | 1   |
| RN1–RN2                     | NOMCA16031002AT1              | R-2R ladder networks                  | 2   |
| U4                          | MCP4921T-E/SN                 | Bench converter                       | 1   |
| U7                          | TC4427AVOA                    | Gate driver                           | 1   |
| Q2–Q5                       | AO3400A                       | Bridge output stage                   | 4   |
| T1                          | SEANERGY-TX-01                | Matching transformer                  | 1   |
| <i>Receive and sensing</i>  |                               |                                       |     |
| D3–D6                       | BAT54S                        | Receive limiter                       | 4   |
| U6                          | OPA1652AIDR                   | Receive amplifier                     | 1   |
| U8                          | ADS1115IDGSR                  | Sensor front end                      | 1   |
| U9                          | INA226AIDGSR                  | Current and energy monitor            | 1   |
| S1–S4                       | Probe set                     | Temperature, TDS, turbidity, pressure | 4   |
| BZ1–BZ2                     | GU1008C-40T/R,<br>PZT-200K-WB | Transducers                           | 2   |
| <i>Mechanical</i>           |                               |                                       |     |
| ENC1                        | SEANERGY-POD-V2               | Enclosure                             | 1   |
| OR1–OR2                     | AS568-246-NBR70               | Face seals                            | 2   |
| PEN1–PEN2                   | WETLINK-M10                   | Cable penetrators                     | 2   |
| H1–H12                      | M3X12-A4-70                   | Closure fasteners                     | 12  |

**Table 5:** Principal line items by subsystem. The complete bill, with supplier references, tolerances, lead times and unit costs, is *01-hardware/bom/bom.csv*: 36 line items covering every reference designator on the board.

| Qty | Component factor | PCB     | Enclosure | Assembly  | Unit cost  |
|-----|------------------|---------|-----------|-----------|------------|
| 1   | 1.00             | INR 800 | INR 1,200 | INR 1,500 | INR 27,648 |
| 10  | 0.82             | INR 500 | INR 800   | INR 650   | INR 21,751 |
| 100 | 0.68             | INR 250 | INR 480   | INR 350   | INR 17,501 |

**Table 6:** Quantity cost model. A planning model built on the quantity-one bill, not a quotation.

| ID   | Test                  | Domain | Raw data                                             |
|------|-----------------------|--------|------------------------------------------------------|
| T-01 | Transducer impedance  | Bench  | T-01_impedance-sweep.csv                             |
| T-02 | Converter spectrum    | Bench  | T-02_dac-spectrum.csv                                |
| T-03 | Reconstruction filter | Bench  | T-03_filter-response.csv                             |
| T-04 | Window sidelobes      | Bench  | T-04_window-sidelobes.csv                            |
| T-05 | Chirp tracking        | Bench  | T-05_chirp-track.csv                                 |
| T-06 | Pulse compression     | Bench  | T-06_pulse-compression.csv                           |
| T-07 | Barker-13             | Bench  | T-07_barker13.csv                                    |
| T-08 | Range accuracy        | Tank   | T-08_range-accuracy.csv                              |
| T-09 | Range resolution      | Tank   | T-09_range-resolution.csv                            |
| T-10 | DMA and CPU idle      | Bench  | T-10_dma-cpu.csv                                     |
| T-11 | Power and energy      | Bench  | T-11_power-states.csv, T-11_ping-energy.csv          |
| T-12 | Synthesis comparison  | Bench  | T-12_dds-vs-naive.csv                                |
| T-13 | Adaptation latency    | Bench  | T-13_adaptation-latency.csv                          |
| T-14 | Sediment adaptation   | Tank   | T-14_sediment-adaptation.csv                         |
| T-15 | Closed loop and noise | Tank   | T-15_snr-convergence.csv,<br>T-15_noise-recovery.csv |
| T-16 | Enclosure immersion   | Tank   | T-16_immersion.csv                                   |

**Table 7:** *Test index. Bench files are under 06-validation/bench-results/raw/, tank files under 06-validation/tank-results/raw/.*

| Function                        | GPIO    | Rationale                                      |
|---------------------------------|---------|------------------------------------------------|
| Temperature probe               | 4       | Preserved from the working prototype           |
| Legacy audio bus                | 5, 6, 7 | Preserved for bench A/B comparison             |
| I <sup>2</sup> C data and clock | 8, 9    | Shared sensor, monitor and display bus         |
| Idle and envelope markers       | 2, 3    | Oscilloscope evidence for <a href="#">T-10</a> |
| Rail gate, transmit enable      | 17, 18  | Power gating and driver safe state             |
| 8-bit converter bus             | 35–42   | Contiguous high GPIO bank                      |
| Converter write, chip select    | 47, 48  | i80 hardware strobes                           |

**Table 8:** *Pin allocation. GPIO 2 and GPIO 3 exist to make processor occupancy and transmit envelope externally measurable rather than inferred; GPIO 18 is the transmit enable line held low in BOOT, READY and FAULT.*

| Command                                | Regenerates                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------|
| <code>npm run check</code>             | The build gate: type check, 85 physics assertions, 33 render tests                                      |
| <code>python run_local.py</code>       | The published dataset, nine notebook figures, run metrics; hashes the result against the committed copy |
| <code>python evaluate.py</code>        | Model metrics and five evaluation figures                                                               |
| <code>python qa.py</code>              | 58 cross-checks that every number quoted in 05-models matches its generated source                      |
| <code>python make-figures.py</code>    | Figures 8, 14, 15, 16 and 17 of the main report, from the validation raw data                           |
| <code>pdflatex technical-report</code> | The report itself, three passes                                                                         |

**Table 9:** *Reproduction. Directories are given in Table 25 of the main report.*